

SIMATS ENGINEERING

CO-4

ASSESSMENT TOOL 1

NAME:M.PAVITHRAN

REG NO:192411014

COURSE NAME: JAVA PROGRAMMING

COURSE CODE: CSA0925

1.JAVA PROGRAM

```
import java.applet.Applet;
import java.awt.Graphics;

public class LifeCycleApplet extends Applet {

    public void init() {
        System.out.println("init() method called");
    }

    public void start() {
        System.out.println("start() method called");
    }

    public void paint(Graphics g) {
        g.drawString("Java Applet Life Cycle", 80, 80);
        g.drawString("init() -> start() -> paint()", 50, 110);
    }

    public void stop() {
        System.out.println("stop() method called");
    }

    public void destroy() {
        System.out.println("destroy() method called");
    }
}
```

2.HTML CODE

```
<html>
```

```
<body>
```

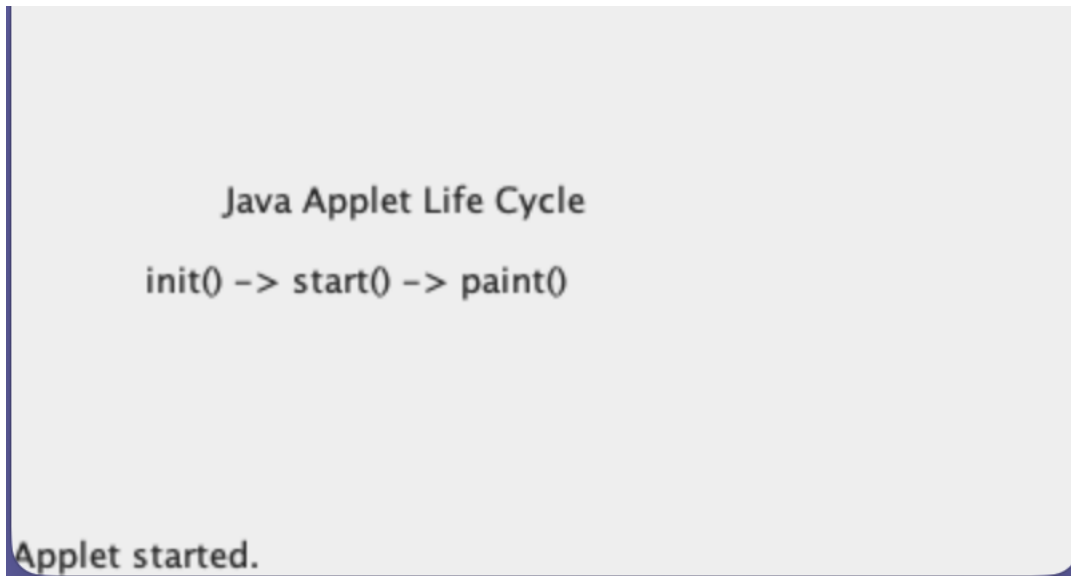
```
<applet code="LifeCycleApplet.class" width="400" height="200">
```

```
</applet>
```

```
</body>
```

```
</html>
```

3.OUTPUT



2.JAVA PROGRAM

```
import java.applet.Applet;
```

```
import java.awt.Color;
```

```
import java.awt.Graphics;
```

```
public class GraphicsApplet extends Applet {
```

```
    public void paint(Graphics g) {
```

```
        g.setColor(Color.CYAN);
```

```
        g.fillRect(0, 0, 500, 300);
```

```
        g.setColor(Color.YELLOW);
```

```
        g.fillOval(380, 30, 70, 70);
```

```
g.setColor(Color.WHITE);  
g.fillOval(70, 50, 70, 40);  
g.fillOval(110, 40, 80, 50);  
g.fillOval(155, 55, 70, 35);
```

```
g.setColor(Color.GREEN);  
g.fillRect(0, 220, 500, 80);
```

```
g.setColor(Color.GRAY);  
g.fillRect(0, 250, 500, 50);
```

```
g.setColor(Color.WHITE);  
for (int x = 20; x < 500; x += 80) {  
    g.fillRect(x, 270, 40, 5);  
}
```

```
g.setColor(Color.ORANGE);  
g.fillRect(180, 150, 150, 100);
```

```
g.setColor(Color.RED);  
int[] xPoints = {160, 255, 350};  
int[] yPoints = {150, 80, 150};  
g.fillPolygon(xPoints, yPoints, 3);
```

```
g.setColor(Color.DARK_GRAY);  
g.fillRect(235, 195, 35, 55);
```

```
g.setColor(Color.WHITE);  
g.fillRect(195, 175, 30, 30);  
g.fillRect(285, 175, 30, 30);
```

```
g.setColor(new Color(139, 69, 19));  
g.fillRect(60, 170, 25, 70);
```

```
g.setColor(Color.GREEN);  
g.fillOval(30, 120, 80, 80);
```

```
g.setColor(new Color(139, 69, 19));  
g.fillRect(400, 170, 25, 70);
```

```
g.setColor(Color.GREEN);
```

```
        g.fillOval(370, 120, 80, 80);  
    }  
}
```

HTML CODE

```
<html>  
<body>  
<applet code="GraphicsApplet.class" width="500" height="300">  
</applet>  
</body>  
</html>
```

OUTPUT:



3.JAVA PROGRAM

```
import java.applet.Applet;  
import java.awt.Graphics;
```

```

public class CalculatorApplet extends Applet {

    public void paint(Graphics g) {
        double a = Double.parseDouble(getParameter("num1"));
        double b = Double.parseDouble(getParameter("num2"));

        double sum = a + b;
        double difference = a - b;
        double product = a * b;
        double quotient = a / b;

        g.drawString("First Number: " + a, 50, 50);
        g.drawString("Second Number: " + b, 50, 75);
        g.drawString("Addition: " + sum, 50, 100);
        g.drawString("Subtraction: " + difference, 50, 125);
        g.drawString("Multiplication: " + product, 50, 150);
        g.drawString("Division: " + quotient, 50, 175);
    }
}

```

HTML CODE

```

<html>
<body>

<applet code="CalculatorApplet.class" width="500" height="250">
<param name="num1" value="20">
<param name="num2" value="5">
</applet>

</body>
</html>

```

OUTPUT:

First Number: 20.0
Second Number: 5.0
Addition: 25.0
Subtraction: 15.0
Multiplication: 100.0
Division: 4.0

Applet started.

4.JAVA CODE

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class StudentProfile extends Frame {
```

```
    Label title, nameLabel, rollLabel, deptLabel, yearLabel, genderLabel, skillsLabel;
```

```
    TextField nameField, rollField, deptField, yearField;
```

```
    Checkbox male, female, java, python, cpp;
```

```
    Button submit, clear;
```

```
    StudentProfile() {
```

```
        setTitle("Student Profile");
```

```
        setSize(500, 450);
```

```
        setLayout(new GridLayout(8, 2, 10, 10));
```

```
        title = new Label("STUDENT PROFILE", Label.CENTER);
```

```
        nameLabel = new Label("Name:");
```

```
        rollLabel = new Label("Roll Number:");
```

```
        deptLabel = new Label("Department:");
```

```
        yearLabel = new Label("Year:");
```

```
        genderLabel = new Label("Gender:");
```

```
        skillsLabel = new Label("Skills:");
```

```
nameField = new TextField();
rollField = new TextField();
deptField = new TextField();
yearField = new TextField();
```

```
Panel genderPanel = new Panel();
male = new Checkbox("Male");
female = new Checkbox("Female");
genderPanel.add(male);
genderPanel.add(female);
```

```
Panel skillsPanel = new Panel();
java = new Checkbox("Java");
python = new Checkbox("Python");
cpp = new Checkbox("C++");
skillsPanel.add(java);
skillsPanel.add(python);
skillsPanel.add(cpp);
```

```
submit = new Button("Submit");
clear = new Button("Clear");
```

```
add(title);
add(new Label(""));
```

```
add(nameLabel);
add(nameField);
```

```
add(rollLabel);
add(rollField);
```

```
add(deptLabel);
add(deptField);
```

```
add(yearLabel);
add(yearField);
```

```
add(genderLabel);
add(genderPanel);
```

```
add(skillsLabel);
```

```
add(skillsPanel);
```

```
add(submit);
```

```
add(clear);
```

```
submit.addActionListener(new ActionListener() {  
    public void actionPerformed(ActionEvent e) {  
        showMessage();  
    }  
});
```

```
clear.addActionListener(new ActionListener() {  
    public void actionPerformed(ActionEvent e) {  
        nameField.setText("");  
        rollField.setText("");  
        deptField.setText("");  
        yearField.setText("");  
        male.setState(false);  
        female.setState(false);  
        java.setState(false);  
        python.setState(false);  
        cpp.setState(false);  
    }  
});
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        dispose();  
    }  
});
```

```
setVisible(true);  
}
```

```
void showMessage() {  
    Dialog d = new Dialog(this, "Student Details", true);  
    d.setLayout(new FlowLayout());
```

```
String gender = male.getState() ? "Male" : female.getState() ? "Female" : "Not  
Selected";
```

```
Label result = new Label(
```

```

        "Name: " + nameField.getText() +
        " | Roll No: " + rollField.getText() +
        " | Department: " + deptField.getText() +
        " | Year: " + yearField.getText() +
        " | Gender: " + gender
    );

    Button ok = new Button("OK");
    ok.addActionListener(e -> d.dispose());

    d.add(result);
    d.add(ok);
    d.setSize(650, 150);
    d.setVisible(true);
}

public static void main(String[] args) {
    new StudentProfile();
}

```

OUTPUT :

The screenshot shows a Java Swing window titled "Student Profile". Inside the window, there is a form titled "STUDENT PROFILE". The form has the following fields and controls:

- Name:** A text field containing "NITHISH".
- Roll Number:** A text field containing "192411016".
- Department:** A text field containing "CSE".
- Year:** A text field containing "3RD".
- Gender:** Two radio buttons, "Male" (checked) and "Female" (unchecked).
- Skills:** Three checkboxes, "Java" (checked), "Python" (checked), and "C++" (checked).
- Buttons:** Two buttons at the bottom, "Submit" and "Clear".

5.JAVA CODE

```

import java.awt.*;
import java.awt.event.*;

```

```
public class Calculator extends Frame implements ActionListener {
```

```
    TextField display;
```

```
    Button[] buttons;
```

```
    String operator = "";
```

```
    double num1 = 0;
```

```
    Calculator() {
```

```
        setTitle("Calculator");
```

```
        setSize(350, 450);
```

```
        setLayout(new BorderLayout());
```

```
        display = new TextField();
```

```
        display.setEditable(false);
```

```
        display.setFont(new Font("Arial", Font.BOLD, 24));
```

```
        add(display, BorderLayout.NORTH);
```

```
        Panel panel = new Panel();
```

```
        panel.setLayout(new GridLayout(4, 4, 5, 5));
```

```
        String[] names = {
```

```
            "7", "8", "9", "/",
```

```
            "4", "5", "6", "*",
```

```
            "1", "2", "3", "-",
```

```
            "0", "C", "=", "+"
```

```
        };
```

```
        buttons = new Button[names.length];
```

```
        for (int i = 0; i < names.length; i++) {
```

```
            buttons[i] = new Button(names[i]);
```

```
            buttons[i].addActionListener(this);
```

```
            panel.add(buttons[i]);
```

```
        }
```

```
        add(panel, BorderLayout.CENTER);
```

```
        addWindowListener(new WindowAdapter() {
```

```
            public void windowClosing(WindowEvent e) {
```

```
                dispose();
```

```

    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();

    if (command.matches("[0-9]")) {
        display.setText(display.getText() + command);
    }
    else if (command.equals("C")) {
        display.setText("");
        num1 = 0;
        operator = "";
    }
    else if (command.equals("=")) {
        double num2 = Double.parseDouble(display.getText());
        double result = 0;

        switch (operator) {
            case "+": result = num1 + num2; break;
            case "-": result = num1 - num2; break;
            case "*": result = num1 * num2; break;
            case "/": result = num1 / num2; break;
        }

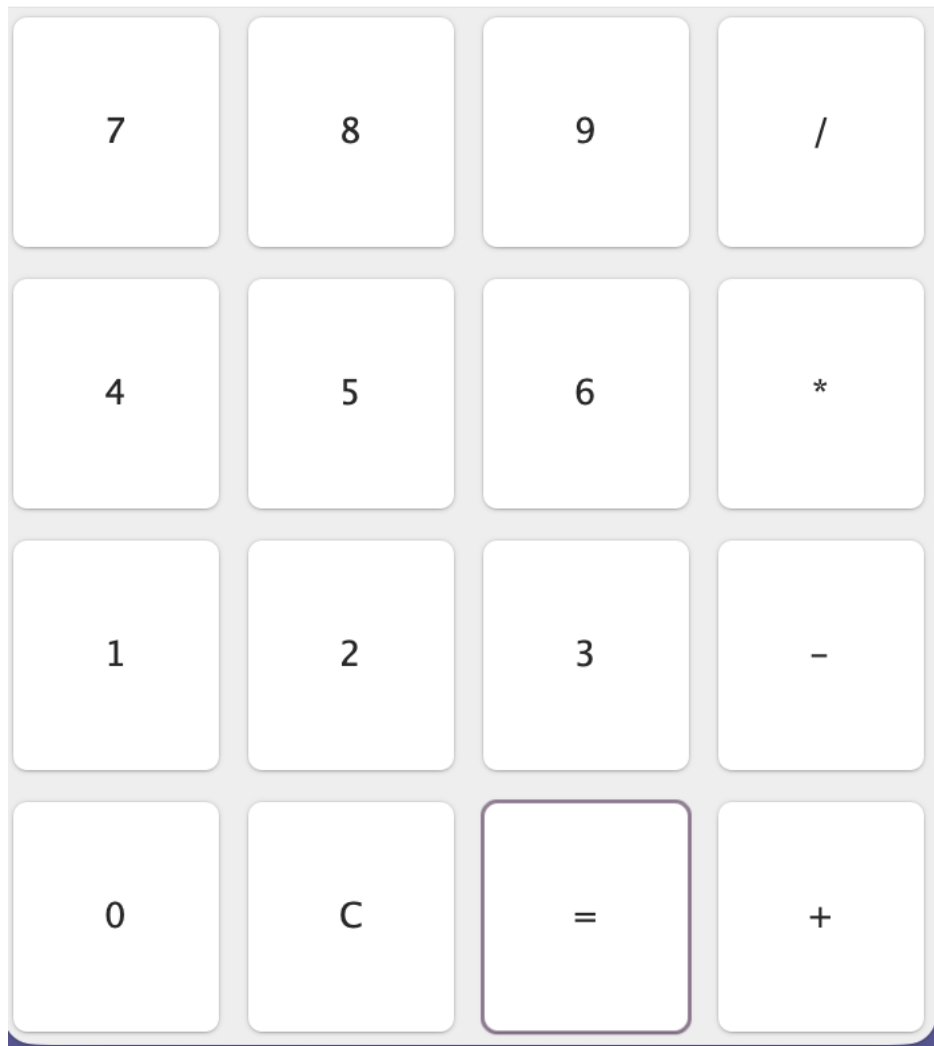
        display.setText(String.valueOf(result));
    }
    else {
        num1 = Double.parseDouble(display.getText());
        operator = command;
        display.setText("");
    }
}

public static void main(String[] args) {
    new Calculator();
}
}

```

OUTPUT:

11.0



6.JAVA CODE:

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class StudentRegistration extends Frame implements ActionListener {
```

Label title, nameLabel, rollLabel, emailLabel, phoneLabel;
Label genderLabel, courseLabel, addressLabel;
TextField nameField, rollField, emailField, phoneField;
Checkbox male, female;
Choice courseChoice;
TextArea addressArea;
Button registerButton, clearButton;

```
StudentRegistration() {  
    setTitle("Student Registration Form");  
    setSize(600, 500);  
    setLayout(new GridBagLayout());  
  
    GridBagConstraints gbc = new GridBagConstraints();  
    gbc.insets = new Insets(8, 8, 8, 8);  
    gbc.fill = GridBagConstraints.HORIZONTAL;  
  
    title = new Label("STUDENT REGISTRATION FORM", Label.CENTER);  
    title.setFont(new Font("Arial", Font.BOLD, 20));  
  
    nameLabel = new Label("Name:");  
    rollLabel = new Label("Roll Number:");  
    emailLabel = new Label("Email:");  
    phoneLabel = new Label("Phone:");  
    genderLabel = new Label("Gender:");  
    courseLabel = new Label("Course:");  
    addressLabel = new Label("Address:");  
  
    nameField = new TextField(25);  
    rollField = new TextField(25);  
    emailField = new TextField(25);  
    phoneField = new TextField(25);  
  
    Panel genderPanel = new Panel();  
    male = new Checkbox("Male");  
    female = new Checkbox("Female");  
    genderPanel.add(male);  
    genderPanel.add(female);  
  
    courseChoice = new Choice();  
    courseChoice.add("CSE");  
}
```

```
courseChoice.add("ECE");
courseChoice.add("EEE");
courseChoice.add("MECH");
courseChoice.add("CIVIL");

addressArea = new TextArea(4, 25);

registerButton = new Button("Register");
clearButton = new Button("Clear");

addComponent(title, gbc, 0, 0, 2, 1);

addComponent(nameLabel, gbc, 0, 1, 1, 1);
addComponent(nameField, gbc, 1, 1, 1, 1);

addComponent(rollLabel, gbc, 0, 2, 1, 1);
addComponent(rollField, gbc, 1, 2, 1, 1);

addComponent(emailLabel, gbc, 0, 3, 1, 1);
addComponent(emailField, gbc, 1, 3, 1, 1);

addComponent(phoneLabel, gbc, 0, 4, 1, 1);
addComponent(phoneField, gbc, 1, 4, 1, 1);

addComponent(genderLabel, gbc, 0, 5, 1, 1);
addComponent(genderPanel, gbc, 1, 5, 1, 1);

addComponent(courseLabel, gbc, 0, 6, 1, 1);
addComponent(courseChoice, gbc, 1, 6, 1, 1);

addComponent(addressLabel, gbc, 0, 7, 1, 1);
addComponent(addressArea, gbc, 1, 7, 1, 1);

addComponent(registerButton, gbc, 0, 8, 1, 1);
addComponent(clearButton, gbc, 1, 8, 1, 1);

registerButton.addActionListener(this);
clearButton.addActionListener(this);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
```

```

        dispose();
    }
});

setVisible(true);
}

void addComponent(Component component, GridBagConstraints gbc,
    int x, int y, int width, int height) {
    gbc.gridx = x;
    gbc.gridy = y;
    gbc.gridwidth = width;
    gbc.gridheight = height;
    add(component, gbc);
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == registerButton) {

        String gender = "";

        if (male.getState())
            gender = "Male";
        else if (female.getState())
            gender = "Female";

        Dialog dialog = new Dialog(this, "Registration Successful", true);
        dialog.setLayout(new FlowLayout());

        Label message = new Label(
            "Student " + nameField.getText() +
            " registered successfully!"
        );

        Button ok = new Button("OK");

        ok.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                dialog.dispose();
            }
        });
    }
}

```

```

    });

    dialog.add(message);
    dialog.add(ok);
    dialog.setSize(400, 150);
    dialog.setVisible(true);

} else if (e.getSource() == clearButton) {

    nameField.setText("");
    rollField.setText("");
    emailField.setText("");
    phoneField.setText("");
    addressArea.setText("");

    male.setState(false);
    female.setState(false);

    courseChoice.select(0);
}
}

public static void main(String[] args) {
    new StudentRegistration();
}
}

```

OUTPUT:

STUDENT REGISTRATION FORM

Name:	Pavithran
Roll Number:	192411016
Email:	Pavithran72@gmail.com
Phone:	
Gender:	☒ Male ☐ Female
Course:	CSE 
Address:	c906,central park
Register Clear	

7 JAVA CODE :

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class BorderLayoutDemo extends Frame {
```

```
    BorderLayoutDemo() {  
        setTitle("BorderLayout Demo");  
        setSize(600, 400);  
        setLayout(new BorderLayout(10, 10));  
  
        Button north = new Button("NORTH");  
        Button south = new Button("SOUTH");  
        Button east = new Button("EAST");  
        Button west = new Button("WEST");  
        Button center = new Button("CENTER");  
  
        add(north, BorderLayout.NORTH);  
        add(south, BorderLayout.SOUTH);
```

```

        add(east, BorderLayout.EAST);
        add(west, BorderLayout.WEST);
        add(center, BorderLayout.CENTER);

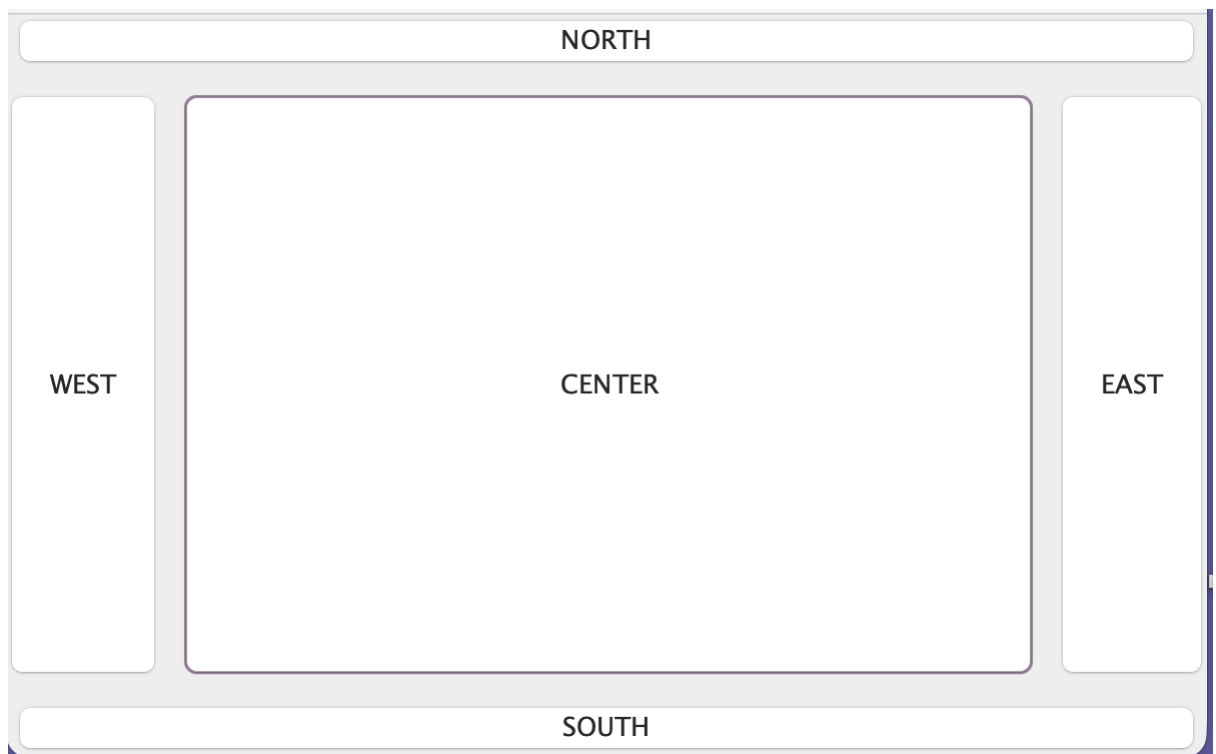
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setVisible(true);
    }

    public static void main(String[] args) {
        new BorderLayoutDemo();
    }
}

```

OUTPUT



8 JAVA CODE

```

import java.awt.*;
import java.awt.event.*;

public class CardLayoutDemo extends Frame implements ActionListener {

```

CardLayout cardLayout;

Panel cardPanel;

Button next, previous;

CardLayoutDemo() {

setTitle("CardLayout Demo");

setSize(500, 350);

setLayout(new BorderLayout());

cardLayout = new CardLayout();

cardPanel = new Panel(cardLayout);

Panel card1 = new Panel();

card1.setLayout(new FlowLayout());

card1.add(new Label("CARD 1 - STUDENT DETAILS"));

card1.add(new Label("Name: Pavithran"));

card1.add(new Label("Department: CSE"));

Panel card2 = new Panel();

card2.setLayout(new FlowLayout());

card2.add(new Label("CARD 2 - COURSE DETAILS"));

card2.add(new Label("Course: Computer Science"));

card2.add(new Label("Year: Second Year"));

Panel card3 = new Panel();

card3.setLayout(new FlowLayout());

card3.add(new Label("CARD 3 - CONFIRMATION"));

card3.add(new Label("Registration Completed"));

cardPanel.add(card1, "Card1");

cardPanel.add(card2, "Card2");

cardPanel.add(card3, "Card3");

Panel buttonPanel = new Panel();

previous = new Button("Previous");

next = new Button("Next");

previous.addActionListener(this);

next.addActionListener(this);

```

        buttonPanel.add(previous);
        buttonPanel.add(next);

        add(cardPanel, BorderLayout.CENTER);
        add(buttonPanel, BorderLayout.SOUTH);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == next) {
            cardLayout.next(cardPanel);
        } else if (e.getSource() == previous) {
            cardLayout.previous(cardPanel);
        }
    }

    public static void main(String[] args) {
        new CardLayoutDemo();
    }
}

```

OUTPUT:

CARD 1 – STUDENT DETAILS Name: Pavithran Department: CSE

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Next